

Course Title: Data Mining and Knowledge Discovery

Project Report: Prediction of Diabetes

Group Number -7

Submitted To – Dr. Faiz Hamid



Sr. No	Name	Roll Number
1)	Amit Gaurav	251140001
2)	Geetanjali	251140008
3)	Krishna Shanmughan	251140011
4)	Manoj Kumar Parida	251140013
5)	Nikhil Madankar	251140014

1.1) Problem Description: Diabetes is one of the most common chronic diseases worldwide, affecting millions of people each year. According to the World Health Organization (WHO), diabetes is a major cause of blindness, kidney failure, heart attacks, stroke, and lower limb amputation. Early detection and prevention play a crucial role in controlling the disease and improving the quality of life of patients.

In this project, the dataset used contains 8 diagnostic features such as number of pregnancies, glucose level, blood pressure, skin thickness, insulin level, body mass index (BMI), diabetes pedigree function, and age. The target variable (Outcome) indicates whether a person is diabetic (1) or non-diabetic (0).

1.2) Data Understanding:

Total Records : 764 Total Attributes : 9

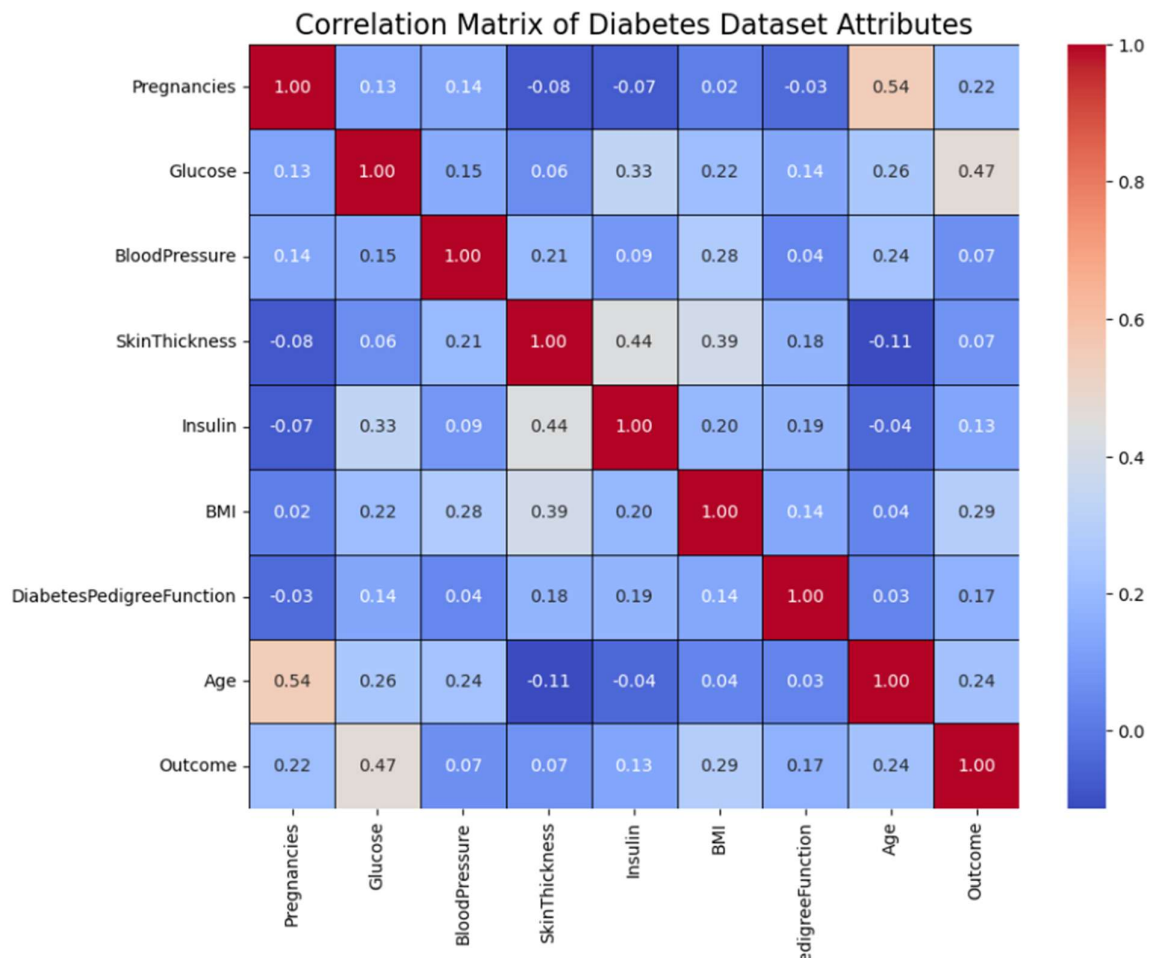
Feature	Description
1) Pregnancies	Number of times pregnant
2) Glucose	Plasma glucose concentration
3) Blood Pressure	Diastolic blood pressure (mm Hg)
4) Skin Thickness	Triceps skin fold thickness (mm)
5) Insulin	2-Hour serum insulin (μ U/mL)
6) BMI	Body Mass Index
7) Diabetes Pedigree Function	Genetic likelihood of diabetes
8) Age	Age in years
9) Outcome	1 = Diabetic, 0 = Non-Diabetic

The description of all numerical attribute using statistical parameter is as follows.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	120.894531	69.105469	20.536458	79.799479	31.992578	0.471876	33.240885	0.348958
std	3.369578	31.972618	19.355807	15.952218	115.244002	7.884160	0.331329	11.760232	0.476951
min	0.000000	0.000000	0.000000	0.000000	0.000000	0.000000	0.078000	21.000000	0.000000
25%	1.000000	99.000000	62.000000	0.000000	0.000000	27.300000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	30.500000	32.000000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

1.3) Exploratory Data Analysis (EDA)

- **Correlation Matrix:** Glucose, BMI, and Age showed the strongest positive correlation with the Outcome variable. Blood Pressure, Skin Thickness, Diabetes Pedigree Function, variables have **low correlation coefficients (below 0.15–0.20)**, meaning their relationship with diabetes outcome is **weak** or **not linear**.



- **Violin and Boxplots:** Indicated higher median glucose and BMI levels in diabetic individuals.

2.0) Data preprocessing:

2.1) Data cleaning: Handling Missing Values Zero values were replaced using the **median** for each attribute.

2.2) Feature Engineering: Added a new feature, Glucose_Insulin_Ratio, to enhance predictive power.

The Glucose_Insulin_Ratio acts as a marker of insulin sensitivity or **insulin resistance**:

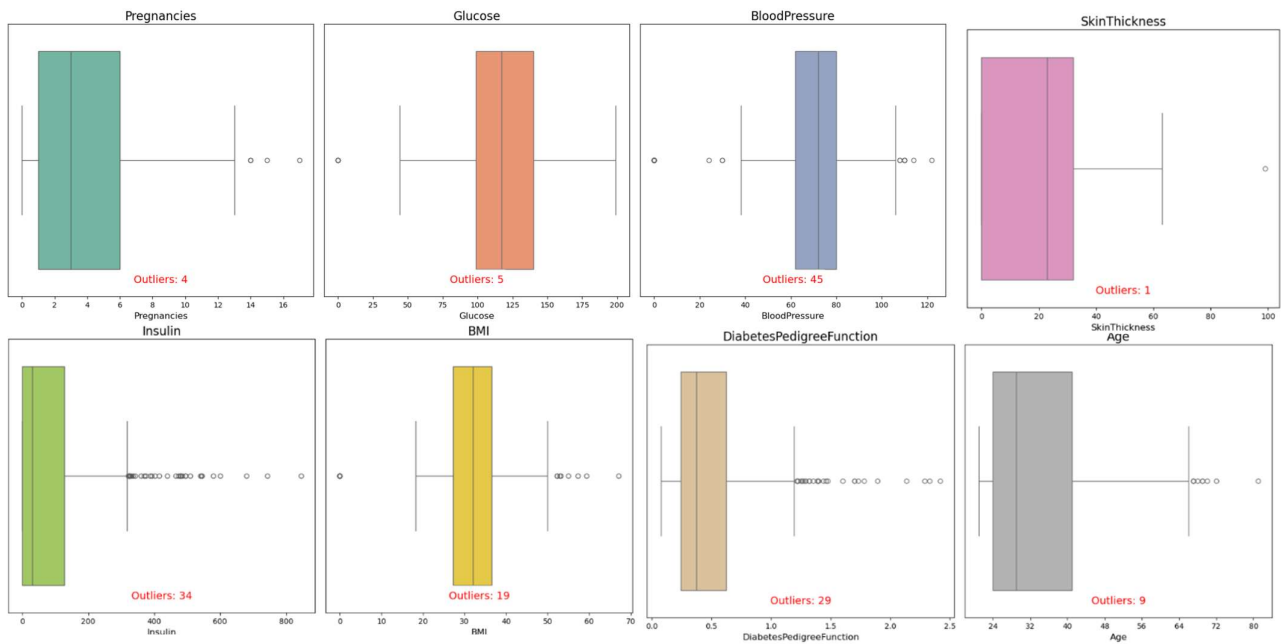
High ratio → Possible insulin resistance / diabetes

Low ratio → Normal insulin response

2.3) Outlier Detection & Treatment

Outliers were detected using **boxplots** and **IQR (Interquartile Range)** method. Values beyond the upper and lower bounds were capped at the boundary limits

- Pregnancies: There were 4 entries of pregnancies which were beyond 14 so we removed those rows
- Glucose, Blood Pressure, BMI : All Outliers values were replaced using the **median**
- Skin Thickness, Insulin: All Outliers & zero values were replaced using the **median**
- Age, Diabetes Pedigree Function: Outliers were not removed as it can be the actual cause diabetes.

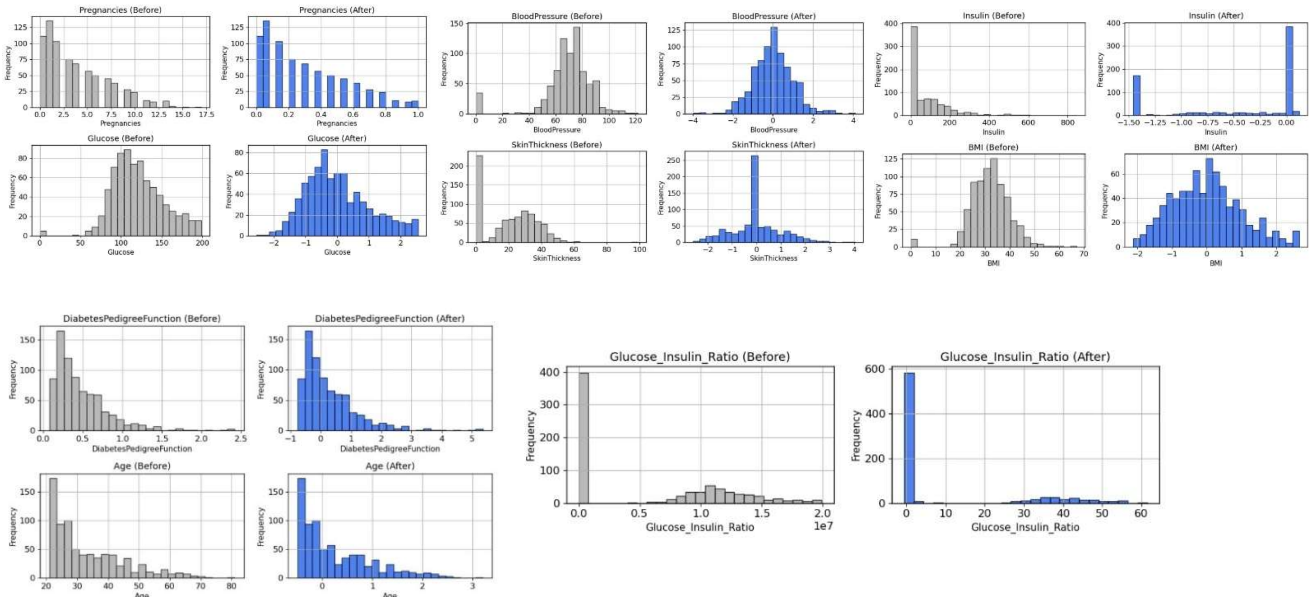


2.4) Normalization

All numerical features were normalized to ensure uniform scale and to improve model performance.

Attribute	Normalization	Reason
1) Pregnancies	Min- max	All outliers treated , squeezing data Does not change the underlying shape of the original distribution, only rescales and centers the data to a common scale
2) Glucose	Z score	
3) Blood Pressure	Z score	
4) Skin Thickness	Z score	
5) BMI	Z score	Retains the original distribution shape of the data while cantering it around zero and scaling it by the spread • Can be applied to datasets that do not follow a normal distribution or have skewed data
6) Insulin	Robust	
7) Diabetes Pedigree Function	Robust	
8) Age	Robust	

Distribution before Normalization & after Normalization



3) Modelling-

Data Splitting:

A) Train-Test- Data Split [70:30]:

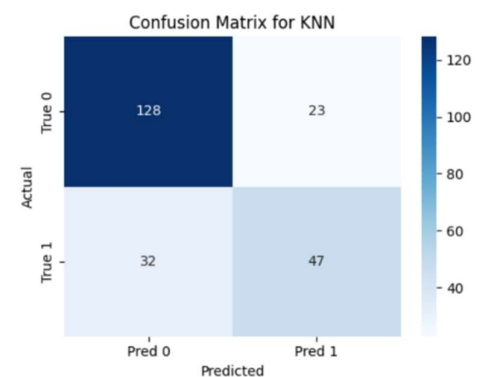
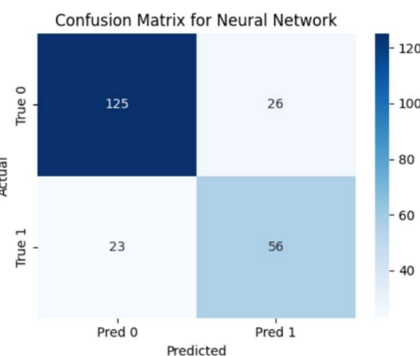
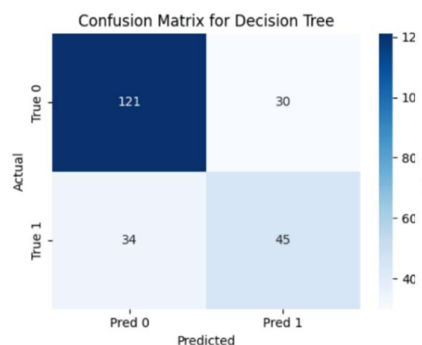
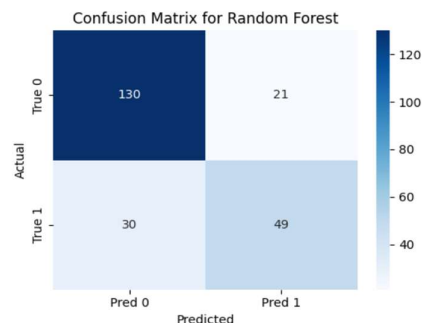
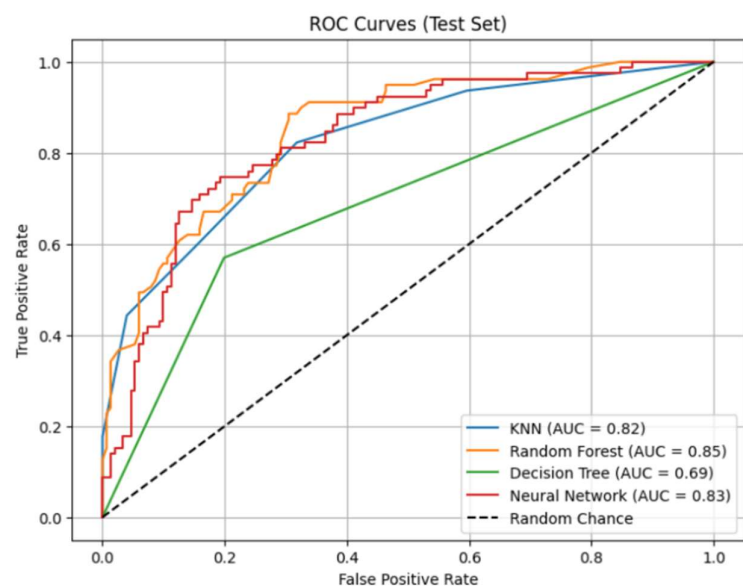
To evaluate the model performance effectively and prevent overfitting, the pre-processed dataset was divided into two subsets using the **train-test split** technique.

The dataset was split in the ratio of **70:30**

KNN	Random Forest	Decision Tree	Neural Network
Accuracy : 0.7609	Accuracy : 0.7783	Accuracy: 0.7217	Accuracy: 0.7870
Precision : 0.6714	Precision : 0.7000	Precision: 0.6000	Precision: 0.6829
Recall : 0.5949	Recall : 0.6203	Recall: 0.5696	Recall: 0.7089
F1 Score : 0.6309	F1 Score : 0.6577	F1 Score: 0.5844	F1 Score: 0.6957
F2 Score : 0.6088	F2 Score : 0.6347	F2 Score: 0.5754	F2 Score: 0.7035
ROC AUC : 0.8240	ROC AUC : 0.8494	ROC AUC: 0.6855	ROC AUC: 0.8322

ROC Curve : To evaluate the classification performance of the developed models, the **Receiver Operating Characteristic (ROC)** curve was plotted for each model.

As shown in the ROC Curve (Figure X), the **Random Forest** model achieved the highest AUC of **0.85**, followed by the Neural Network (AUC = 0.83) and KNN (AUC = 0.82), demonstrating good classification capability.



B) Train-Test- Data split [60:20:20]:

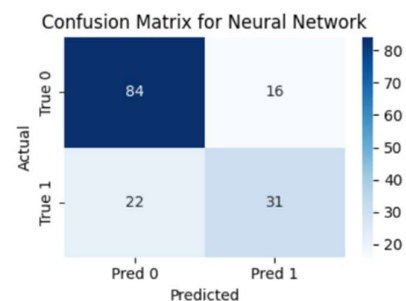
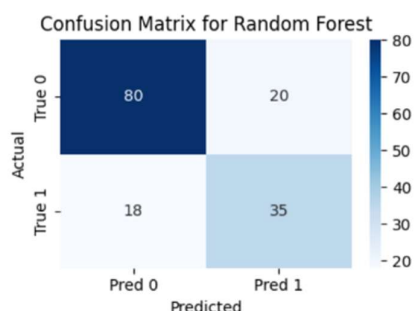
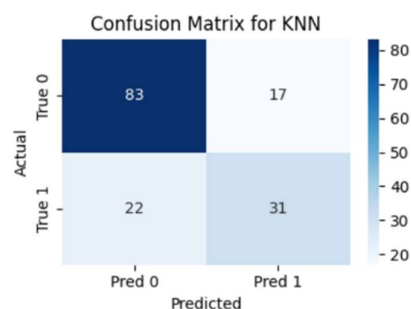
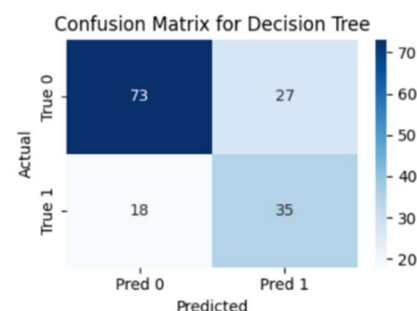
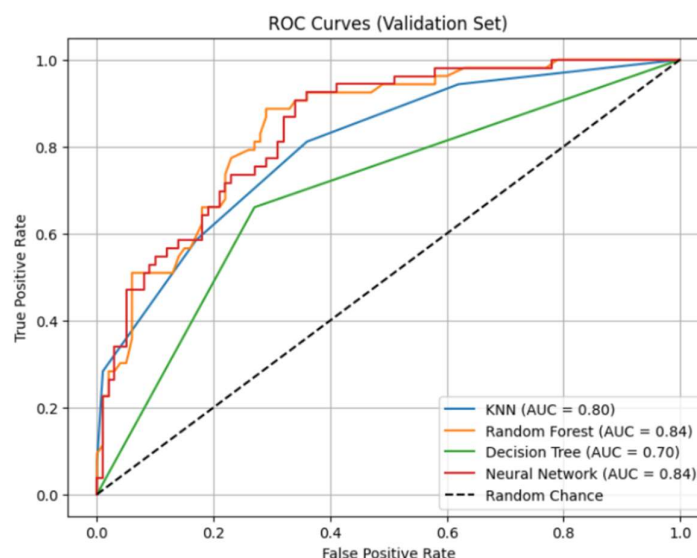
The processed dataset was divided into three subsets: **60% for training, 20% for validation, and 20% for testing**. To assess the performance of different machine learning models, the processed dataset was divided into **three subsets: 60% for training, 20% for validation, and 20% for testing**.

Four classification algorithms — **K-Nearest Neighbors (KNN)**, **Random Forest**, **Decision Tree**, and **Neural Network** — were trained and evaluated on the same split.

The table below summarizes their performance across key evaluation metrics such as **Accuracy**, **Precision**, **Recall**, **F1 Score**, and **ROC-AUC**.

KNN	Random Forest	Decision Tree	Neural Network
Accuracy : 0.7451	Accuracy : 0.7516	Accuracy: 0.7059	Accuracy: 0.7516
Precision : 0.6458	Precision : 0.6364	Precision: 0.5645	Precision: 0.6596
Recall : 0.5849	Recall : 0.6604	Recall: 0.6604	Recall: 0.5849
F1 Score : 0.6139	F1 Score : 0.6481	F1 Score: 0.6087	F1 Score: 0.6200
F2 Score : 0.5962	F2 Score : 0.6554	F2 Score: 0.6387	F2 Score: 0.5985
ROC AUC : 0.8012	ROC AUC : 0.8449	ROC AUC: 0.6952	ROC AUC: 0.8438

- Among the evaluated models, the **Random Forest** and Neural Network achieved the highest AUC values (0.84), indicating excellent classification performance.
- The KNN model also performed well with an AUC of **0.80**, while the Decision Tree model showed the lowest AUC (0.70), suggesting it was less effective on the validation data.
- The results confirm that ensemble-based models (Random Forest) and deep learning approaches (Neural Network) generalize better than simpler models such as Decision Tree and KNN.



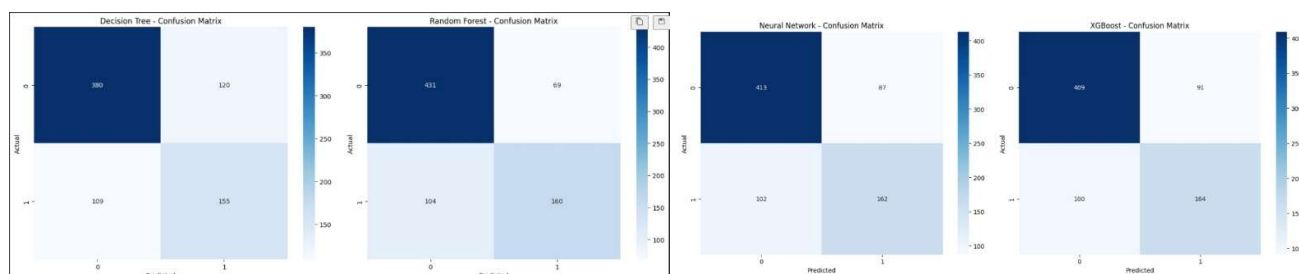
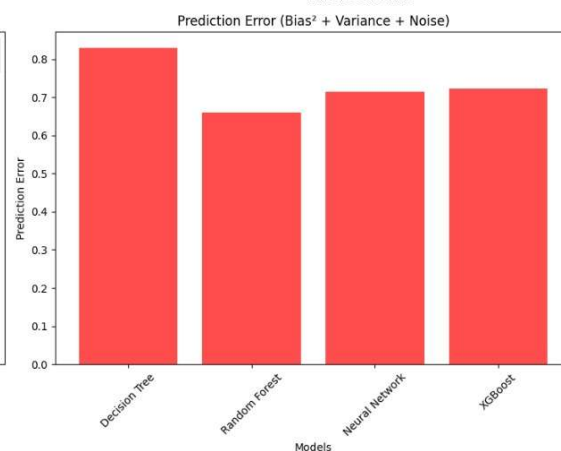
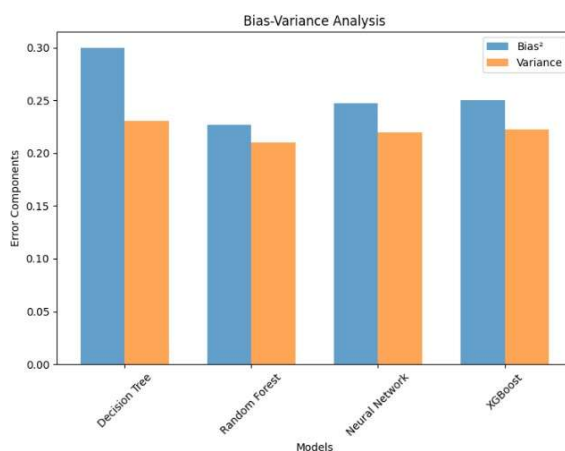
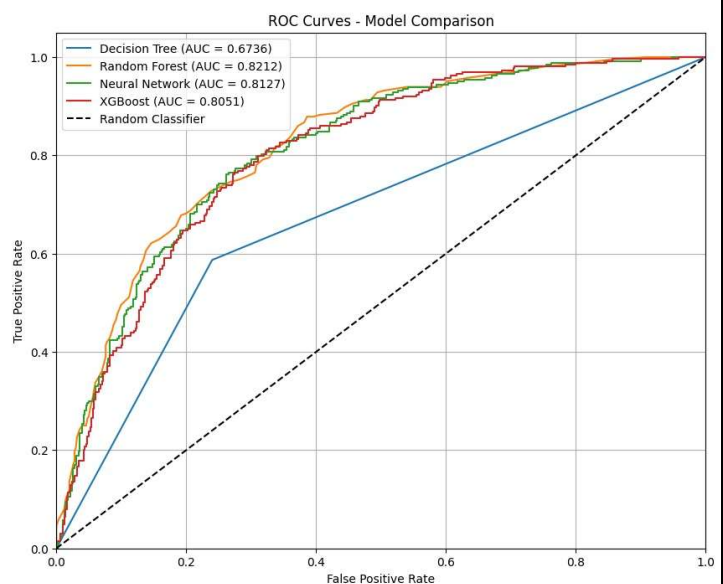
C) K-Fold Cross Validation (Leave One Out):

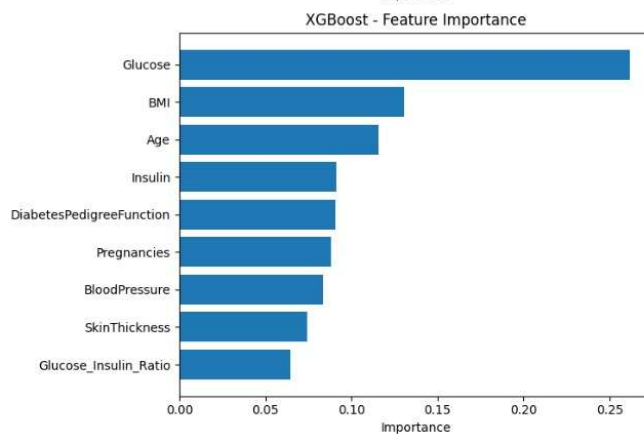
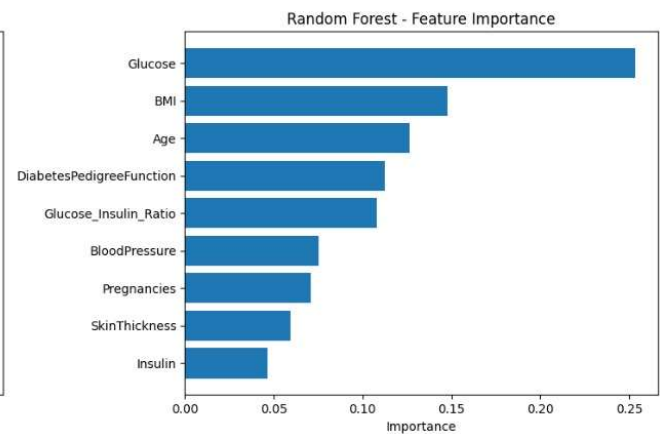
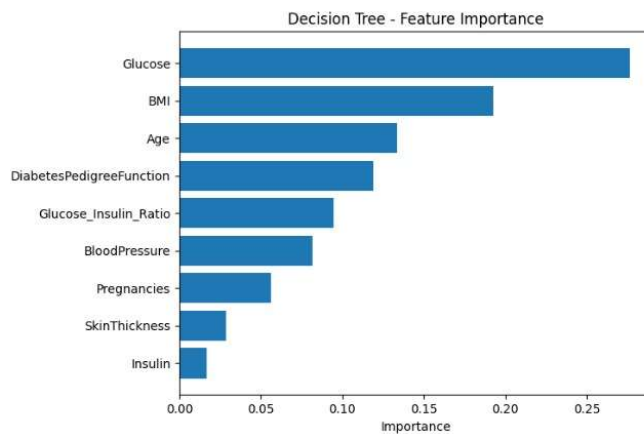
To ensure the robustness and generalization capability of the developed models, K-Fold Cross Validation was performed using the **Leave-One-Out (LOO)** approach.

In this method, each observation in the dataset is used once as a validation sample, while all remaining observations are used for training.

Decision Tree	Random Forest	Neural Network	XGBoost	Ensemble
Accuracy : 0.7003	Accuracy : 0.7736	Accuracy : 0.7526	Accuracy : 0.7500	Accuracy : 0.7749
Precision : 0.5636	Precision : 0.6987	Precision : 0.6506	Precision : 0.6431	Precision : 0.6901
Recall : 0.5871	Recall : 0.6061	Recall : 0.6136	Recall : 0.6212	Recall : 0.6326
Specificity : 0.7600	Specificity : 0.8620	Specificity : 0.8240	Specificity : 0.8160	Specificity : -
F1 Score : 0.5751	F1 Score : 0.6491	F1 Score : 0.6316	F1 Score : 0.6320	F1 Score : 0.6601
F2 Score : 0.5823	F2 Score : 0.6226	F2 Score : 0.6207	F2 Score : 0.6255	F2 Score : 0.6433
ROC AUC : 0.6736	ROC AUC : 0.8212	ROC AUC : 0.8127	ROC AUC : 0.8051	ROC AUC : -

- The Leave-One-Out Cross Validation confirmed that the **Random Forest** model achieved the highest overall performance (Accuracy = 77.36%, **ROC-AUC = 0.8212**), followed by Neural Network (ROC-AUC = 0.8127) and XGBoost (ROC-AUC = 0.8021).
- These results validate the superior generalization capability of ensemble and neural models over traditional decision trees in predicting diabetes outcomes.





Contribution:

Name	Contribution:
1) Krishna Shanmughan (251140011)	- Exploratory Data Analysis (EDA) - Helped in improving the accuracy, F-Beta Score, F-1 score, Recall, Precision of best model i.e. Ensemble model from XG Boost - Report Writing
2) Nikhil Madankar (251140014)	- Data preprocessing (Data Cleaning , Outlier Handling & Normalization) - Feature Engineering - Report Writing
3) Geetanjali (251140008)	- Train-Test- Data Split [70:30] - Models : KNN, Random Forest , Decision Tree, Neural Network
4) Manoj Kumar Parida (251140013)	- Train-Validation-Test- Data Split [60:20:20] - Models : KNN, Random Forest , Decision Tree, Neural Network
5) Amit Gaurav (251140001)	- K-Fold Cross Validation (Leave One Out): - Models : Random Forest , Decision Tree, XG Boost, Neural Network - Improving the accuracy, F-Beta Score, F-1 score, Recall, Precision of best model i.e. Ensemble model from XG Boost